

Multiple Choice (10 marks)

1. What is the second most common element in the solar system?
 - (a) Iron
 - (b) Hydrogen
 - (c) Methane
 - (d) Helium
2. Which is the least likely cause of death?
 - (a) Being hit in the head by a bullet.
 - (b) Being hit by a small meteorite.
 - (c) Starvation during global winter caused by a major impact.
 - (d) Driving while intoxicated without wearing seatbelts.
3. How do we know how old the Earth is?
 - (a) From the layering of materials within the Earth.
 - (b) From fossils of ancient life.
 - (c) From the cratering history of Earth's surface.
 - (d) From radioactive dating of rocks and meteorites.
4. The lithosphere of a planet is the layer that consists of
 - (a) the softer rocky material of the mantle.
 - (b) the lava that comes out of volcanoes.
 - (c) material between the crust and the mantle.
 - (d) the rigid rocky material of the crust and uppermost portion of the mantle.
5. Which property of our solar system does a formation theory NOT need to explain?
 - (a) patterns of motion
 - (b) the presence of asteroids and comets
 - (c) the measured ages of meteorites
 - (d) the presence of life on Earth

True / False (10 marks)

Answer True or False

6. True or False: One astronomical unit (AU) is defined as approximately 150 million kilometers from the Earth to the Sun.
7. True or False: Venus experiences no seasons because it does not rotate fast enough to create significant temperature variations.
8. True or False: The Pleiades is an open star cluster that is commonly recognized for containing seven prominent bright stars.
9. True or False: Ganymede is the closest of Jupiter's four largest moons, orbiting at a smaller distance than Io, Europa, and Callisto.
10. True or False: The rock labeled as 'bigfoot' on Mars was about 10 meters away from the rover, which is a common misinterpretation.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the reasons behind Saturn's comparable size to Jupiter despite its lower mass, focusing on the implications of density and gravitational compression on planetary structure.
12. Discuss the relationship between mass and weight, and analyze how the weight of a 5 kg sandbag would change on a planet with half the gravitational pull of Earth.

End of Paper